

A Literature-Implied Bound for the Uniformly Discrete $p = 1$ Strong-Schur Route

Math discovery smoke-test packet

June 12, 2026

1 Status

This packet records a *literature-implied* answer, not a new original pipeline proof. The source paper is

F. Albiac, J. L. Ansorena, J. Bima, and M. Cuth, *Lipschitz free p -spaces for $0 < p < 1$ in the light of the Schur p -property and the compact reduction*, arXiv:2506.09786.

The supporting paper is

M. Cuth and O. F. K. Kalenda, *Lipschitz-free spaces over uniformly discrete metric spaces are 3-Schur*, arXiv:2604.01875.

The supporting paper does not appear to explicitly present itself as answering Question 6.3 of arXiv:2506.09786. The answer below is therefore filed under `literature_implied_answers`.

2 Where the Question Appears

The source paper asks whether the Schur p -property and strong Schur p -property differ for Lipschitz free p -spaces. It then reformulates the problem in terms of uniformly bad K -strong Schur constants and singles out the uniformly separated bounded case.

of Question 6.2, we ask about the existence of p -Schur spaces whose dual space is null.

It will also be interesting to determine whether the Schur p -property and the strong Schur p -property are different in the setting of Lipschitz free p -spaces.

Question 6.3. Let $0 < p \leq 1$. Is there a metric space $\mathcal{M}$ such that $\mathcal{F}_p(\mathcal{M})$ has the Schur p -property but fails to have the strong Schur p -property?

Any Lipschitz free p -space over a metric space is isomorphic to a Lipschitz free p -space over a bounded metric space [5], and the techniques from [4] allow to construct a metric space $\mathcal{M}$ from a countable family $(\mathcal{M}_i)_{i \in \mathbb{I}}$ of bounded metric spaces such that $\mathcal{F}_p(\mathcal{M})$ is isomorphic to $(\bigoplus_{i \in \mathbb{I}} \mathcal{F}_p(\mathcal{M}_i))_{\ell_p}$.

Hence, Question 6.3 is equivalent to asking about the existence for each $K \in [1, \infty)$ of a bounded metric space $\mathcal{M}$ such that $\mathcal{F}_p(\mathcal{M})$ fails to have the K -strong Schur p -property. In particular, we wonder if for each $K \in [1, \infty)$ there is a uniformly separated bounded metric space $\mathcal{M}$ such $\mathcal{F}_p(\mathcal{M})$ fails to

have the K -strong Schur p -property despite being isomorphic to ℓ_p (see [3, Theorem 4.14]).

We note that very recently Kalenda ([18]) found an example of a uniformly separated bounded metric space $\mathcal{M}$ such that $\mathcal{F}(\mathcal{M})$ does not have 1-strong Schur property, see [18, Example 8.5]. However, the techniques to prove this are using duality techniques, so we do not know whether an analogue for $p < 1$ may be proved as well.

In connection with the compact reduction principle discussed in Section 5, we pose several further related questions. For each of the questions above, a positive answer is known when $n = 1$. However, the existing proofs rely on

3 Claim

Theorem 1. *Let M be a uniformly separated bounded metric space. For $p = 1$, the real Lipschitz-free space $\mathcal{F}(M)$ has the 3-strong Schur property. Consequently, the uniformly separated bounded $p = 1$ version of the “for each K ” formulation following Question 6.3 has a negative answer: no such M can fail the K -strong Schur property for every K .*

Proof. For ordinary metric spaces, “uniformly separated” is the same condition as “uniformly discrete”: there is $\delta > 0$ such that distinct points are at least δ apart.

Cuth–Kalenda prove the following theorem.

More precisely, there is a uniformly discrete metric space M (arising from a graph using the shortest-path metric) which is 3-Schur and not c -Schur for any $c < 3$. In the present paper we prove that the constant 3 is optimal. More precisely, the main result is the following:

Theorem 1.1. *Let M be a uniformly discrete metric space. Then $\mathcal{F}(M)$ is 3-Schur. Moreover, the constant 3 is optimal.*

Let us note that our Theorem 1.1 contributes to the recent line of research focused on Banach space theoretical properties of Lipschitz-free spaces over uniformly discrete spaces (hereafter referred to as UDD spaces). This line of research was

Thus, for every uniformly separated metric space M , the real Lipschitz-free space $\mathcal{F}(M)$ is 3-Schur. The same paper recalls the comparison between quantitative Schur and strong Schur constants:

$$\tilde{\delta}(x_n) = \inf\{\delta(x_{n_k}); (n_k) \text{ strictly increasing}\}.$$

The following proposition follows from [21, Proposition 5.1 and Proposition 5.6].

Proposition 2.2. *Let X be a real Banach space and let $c \geq 1$.*

- (a) *The following assertions are equivalent:*
 - (i) *X has the c -strong Schur property;*
 - (ii) *$\tilde{c}\alpha(x_n) \leq c\tilde{\delta}(x_n)$ for any bounded sequence (x_n) in X ;*
 - (iii) *$\tilde{c}\alpha(x_n) \leq c\tilde{\delta}(x_n)$ for any bounded sequence (x_n) in X .*
- (b) *If X is c -Schur, it has also the c -strong Schur property.*
- (c) *If X has the c -strong Schur property, it is $(2c+1)$ -Schur.*

We stress that the previous proposition holds only for real Banach spaces. In

Since Lipschitz-free spaces in that paper are real Banach spaces, Proposition 2.2 applies with $c = 3$. Hence $\mathcal{F}(M)$ has the 3-strong Schur property. In particular, $\mathcal{F}(M)$ cannot fail K -strong Schur for $K = 3$, nor for any larger K , because larger constants give weaker requirements. This contradicts the proposed uniformly separated bounded “for each K ” route in the $p = 1$ case. $\square$

4 Interpretation

This does not solve Question 6.3 as stated. It only closes a natural subroute:

- it is restricted to $p = 1$;
- it is restricted to uniformly separated, hence uniformly discrete, metric spaces;
- it uses a later theorem rather than an original proof found in this run.

The statement is still useful for triage. The source paper notes that Kalenda had produced a uniformly separated bounded metric space whose $\mathcal{F}(M)$ is not 1-strong Schur. The Cuth–Kalenda theorem shows that this phenomenon cannot be amplified into examples with arbitrarily large strong-Schur constants in the real $p = 1$ uniformly discrete setting: the universal constant 3 is already available.

5 Verification

The script `code/check_implication.py` checks that the source and supporting PDFs contain the exact snippets used in this implication:

```
conda run --no-capture-output -n sandbox python code/check_implication.py
```

The run used for this packet returned:

Verified source question text and supporting theorem snippets.

The script `code/crop_evidence.py` regenerates the stitched source question crop and the two supporting theorem/proposition crops from the local PDF files.

6 Human Review Recommendation

Reviewers should check the constant convention in the source paper against the c -strong Schur convention in arXiv:2604.01875. The source’s $p = 1$ definition agrees with the usual strong Schur convention used in the supporting paper, but this is the main semantic hinge. The packet deliberately does not claim anything for $0 < p < 1$.